

DESIGN, FABRICATION & COMPARATIVE STUDY OF HYDROPONIC SYSTEM



Project Guide:

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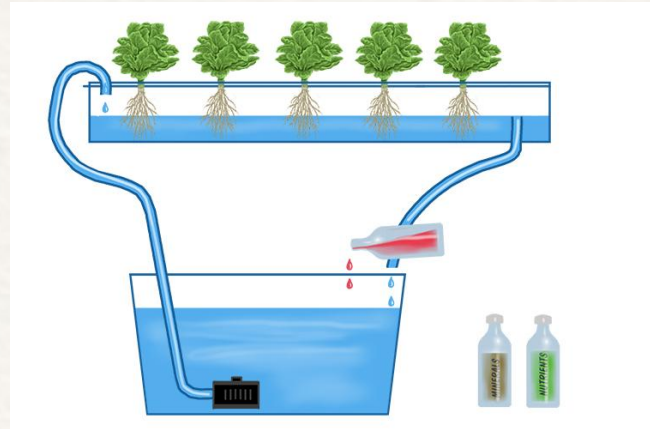
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References



INTRODUCTION

- Sustainable farming product for people who **lack**
 - arable land
 - space
 - water for agriculture
- The project deals with
 - **manufacturing** our own hydroponics system
 - **Comparison** with an existing A- frame hydroponic system
- Hydroponics is basically a method of growing plants without soil by using nutrient solution in an aqueous solvent.



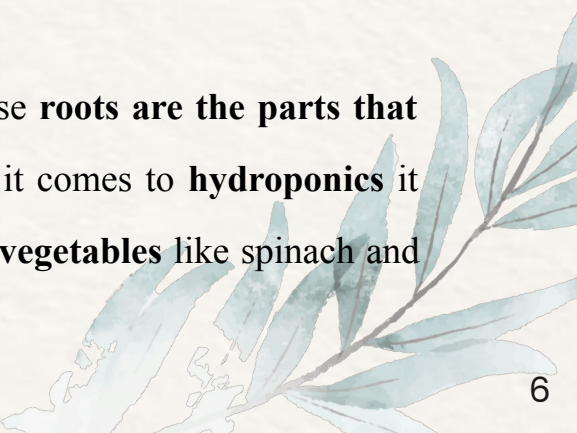
LITERATURE REVIEW

SL. NO	TITLE	JOURNAL	YEAR	AUTHOR AFFILIATION	REMARKS
1	Evaluation of hydroponic systems for the cultivation of Lettuce	Agricultural Water Management	2021	Majid, M., Khan, J.N., Shah, Q.M.A., Masoodi, K.Z., Afroza, B. and Parvaze, College of Agricultural Engineering & Technology, Sher-e-Kashmir University of Agricultural Sciences & Technology of Kashmir, Shalimar Campus, J&K, India	• Deep water culture and NFT(Nutrient Film Technique) provide superior yield and faster growth compared to soil • Deep water culture is easy to use and cheap. NFT is water efficient.
2	Modern Practices for the Cultivation of Leaf Vegetables in Hydroponics	Atlantis Press	2020	Kling, A., Pashek, V., Kumpan, V. and Konoplyov, Department of Horticulture, Forestry and Plant Protection Omsk State Agrarian University	• Plants like lettuce, spinach and basil require less time for harvesting in vertical hydroponics compared to horizontal hydroponics

SL. NO	TITLE	JOURNAL	YEAR	AUTHOR AFFILIATION	REMARKS
3	Vertical farming increases lettuce yield per unit area compared to conventional horizontal hydroponics	Food and Energy Security	2016	Touliatos, D., Dodd, I.C. and McAinsh, The Lancaster Environment Centre, Lancaster University, Lancaster, UK	Vertical hydroponics system can produce about 13.8 times more crop compared to horizontal hydroponics systems.
4	Growth Responses and Root Characteristics of Lettuce Grown in Aeroponics, Hydroponics, and Substrate Culture	Horticulturae	2018	Li, Q., Li, X., Tang, B. and Gu, Department of Horticultural Sciences, Texas A&M AgriLife Extension Service, Texas A & M University, College Station, TX 77843, USA	Lettuce in aeroponics had a greater root/shoot ratio while in hydroponics it had a higher shoot length than the root.



GAPS IDENTIFIED

- In horizontal farming methods **crops per unit area** is very less.
 - Compared to other countries, hydroponic vertical farming with barrels is **not widely practiced** in India [7]
 - The yield in horizontal farming systems is **13.8 times less** compared to vertical farming systems[3]
 - It was found that leafy crops in aeroponics had a greater root/shoot ratio while in hydroponics it had a higher shoot length than the root.
 - This implies that **aeroponics** is best suited for plants whose **roots are the parts that can be consumed** and not the shoot or leaves. But when it comes to **hydroponics** it was identified that it is best suited for the growth of **leafy vegetables** like spinach and lettuce.
- 

MOTIVATION

- Growing population
- Food demand
- Water Scarcity
- Arable land loss
- Most of the existing hydroponics systems are bulky and space consuming



Source :- <https://images.app.goo.gl/HHqaE4zXCKysJQDh7>

OBJECTIVES

- Design of hydroponic vertical barrel in SolidWorks
- Fabrication of the vertical barrel by 3D printing
- Comparative study of hydroponic vertical barrel with that of A-frame system



Methodology

01



Identified the gaps

Designed first product



02

03

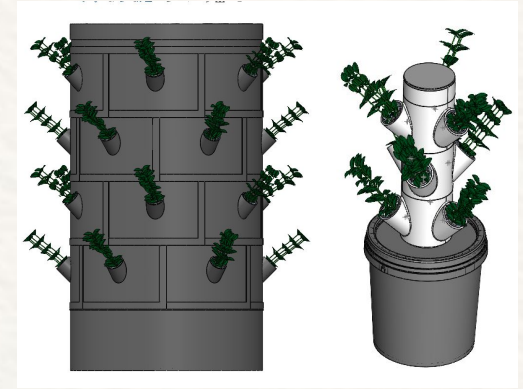


Cost estimation of first product

Designed second product
for 3D Printing



04





05



Cost estimation of second product

3D Printing of Second Product



06



07



Acquisition of the A-Frame farming system from Plantme

Setting up the products



08





09



Germinating the seedlings



Transferring the germinated sprouts to the corresponding systems



10



11



Growing the plants and monitoring



1st yield



12

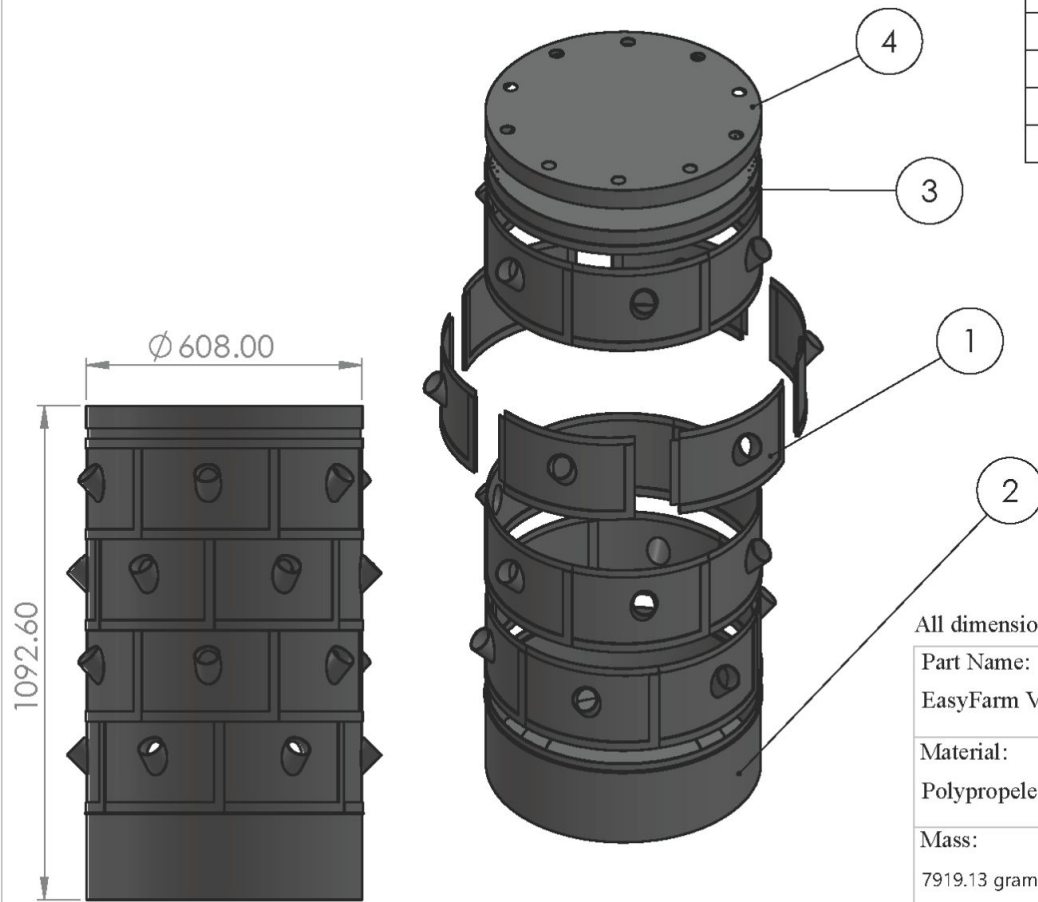


Materials

EasyFarm Vertical Barrel

- Material Used: -
Polypropylene copolymer
- Does not absorb moisture
- Good chemical resistance
- The Product consists of the following **4 parts**:
 - Segment
 - Bucket
 - Buffer
 - Cap
- Initial Capital required 20 Lakhs.

Physical	ASTM Test Method	Homo-Polymer	Co-Polymer
Density (lb/in ³)	D792	0.033	0.033
Specific Gravity	D792	0.905	0.897
Water Absorption @ 24 hours (%)	D570	0.01	-
Mechanical			
Tensile Strength (PSI)	D638	4,800	4800
Tensile Modulus (PSI)	D638	195,000	-
Tensile Elongation at Yield (%)	D638	12	23
Flexural Strength (PSI)	D790	7,000	5400
Flexural Modulus (PSI)	D790	180,000	160000
Thermal			
Melting Point, °F	D3418	327	327

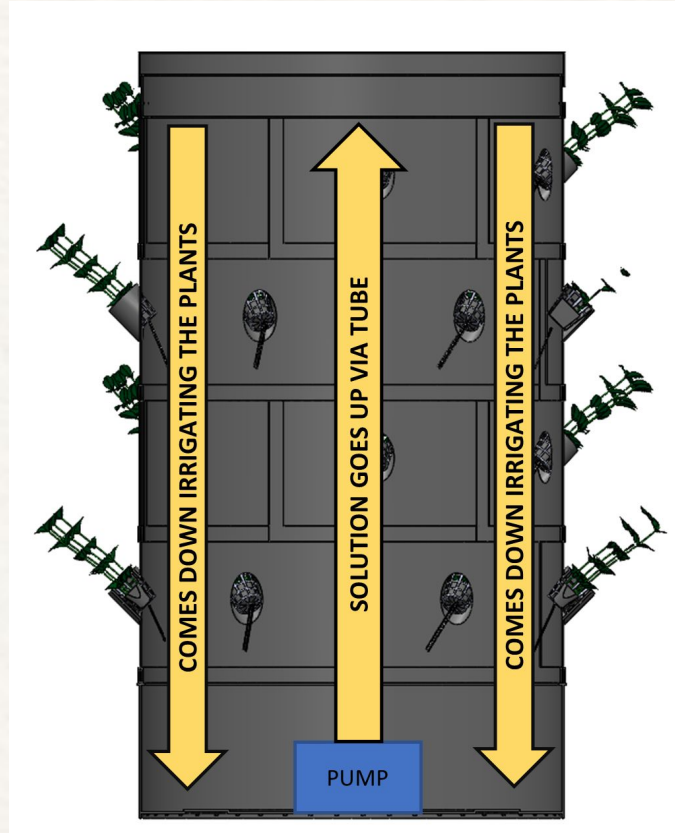


PART NO.	PART NAME	QTY.
1	Segment	24
2	Bucket	1
3	Buffer	1
4	Cap	1

All dimensions are in mm

Part Name: EasyFarm Vertical Barrel Assembly	Date: 28-05-2022
Material: Polypropelene Co-Polymer	Scale: 1:15
Mass: 7919.13 grams	Part no: -

Working



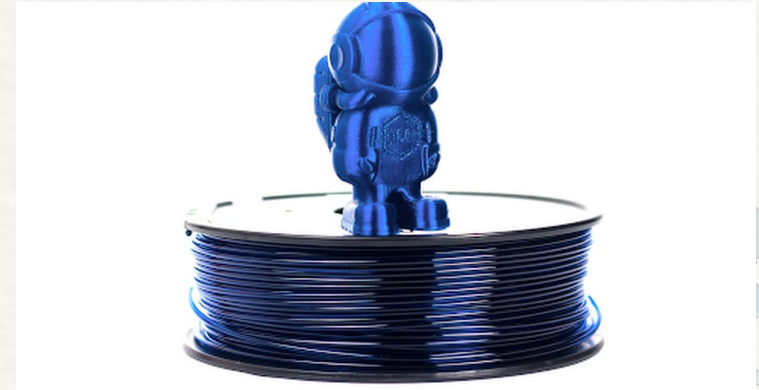
3D Printed Hydroponic Vertical Barrel

- The 3D Printable design consists of 3 parts which needs to be 3D printed
 - Lid
 - Buffer
 - Ring
- 2 parts which is readily available in the market.
 - Bucket
 - Bucket lid
- For injection Moulding 6 lakhs



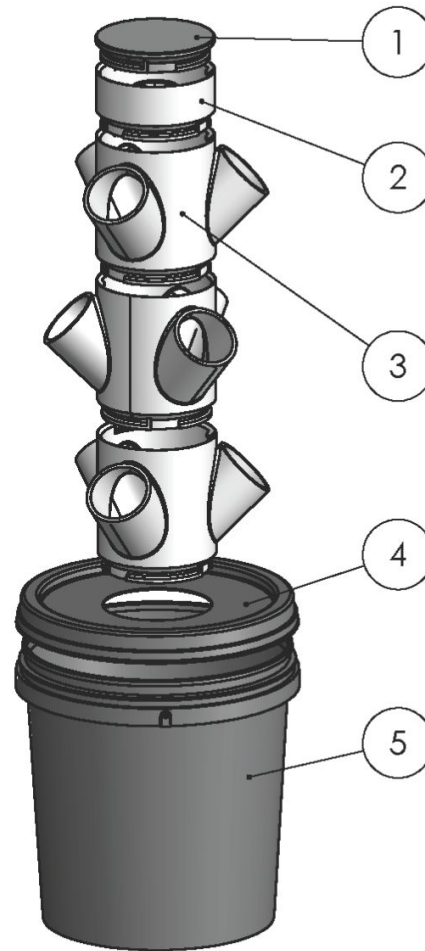
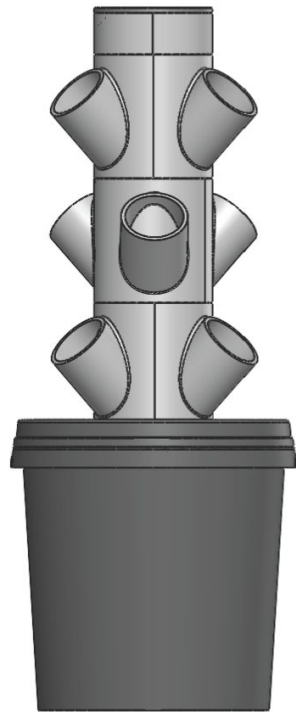
Material Selection

- Selected material is **PETG**
- To create PETG, glycol is added to PET (Polyethylene Terephthalate) at molecular level
- 100% Recyclable
- Greater strength and durability
- food - grade material
- PETG is water, chemical and fatigue resistant
- Melting point : 80°C
- Density : 1.23g/cm³



Source:

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PART NO.	PART NAME	QTY.
1	LID	1
2	BUFFER	1
3	RING	3
4	BUCKET LID	1
5	BUCKET	1

All dimensions are in mm

Part Name: 3D Printed vertical barrel	Date: 30-05-2022
Material: PETG	Scale: 1:7
Mass: 1451.65 grams	Part no: -

Nutrient Solution

- We are using the nutrient solution manufactured by PlantMe

TDS Meter

- Used to find the Total Dissolved Solids
- Helps in monitoring whether enough nutrients are available in the nutrient solution

pH Meter

- Used to find the pH of the nutrient solution
- We have to maintain a pH level between 5.5 to 7
- Low pH can indicate magnesium deficiency or Iron and Aluminium toxicity
- High pH levels can prevent uptake of Calcium and Phosphorus

Site Selection

- We placed our farming equipments in the roof of M George Block at Muthoot Institute of Technology and Science

Seedling Production

- We purchased the seeds from Plantme Aluva
- We germinated it at the college before transferring it into the system

Plant Selected

- Spinach



Cost (₹)

3D Printed vertical barrel	13000
A-frame system	10730
Nutrient Solution	$325 \times 2 = 650$
Pump	$644 \times 2 = 1288$
pH Meter	1900
Green amaranthus seeds	20
Clay pellets	100
Reservoir	180
Hose	50
Total cost	27918

Growing Systems



Vertical Barrel Setup



A Frame Setup

RESULTS AND DISCUSSION

1.Crop performance

- Crop performance was monitored by planting the same crop in two different systems with the same working technique.
- The growing system had a significant influence in the crops performance
- The parameters taken in to consideration are crop duration, crop growth, crop quality & yield.

1.1 Crop Duration

- Crop duration based on the days to transplanting, days to maturity and harvesting were noted.

System	Days to transplant	Days to maturity	Days to harvest
A-frame system	7	14	17
Vertical Barrel system	7	15	17



Plants at mature stage



Plants at harvesting stage

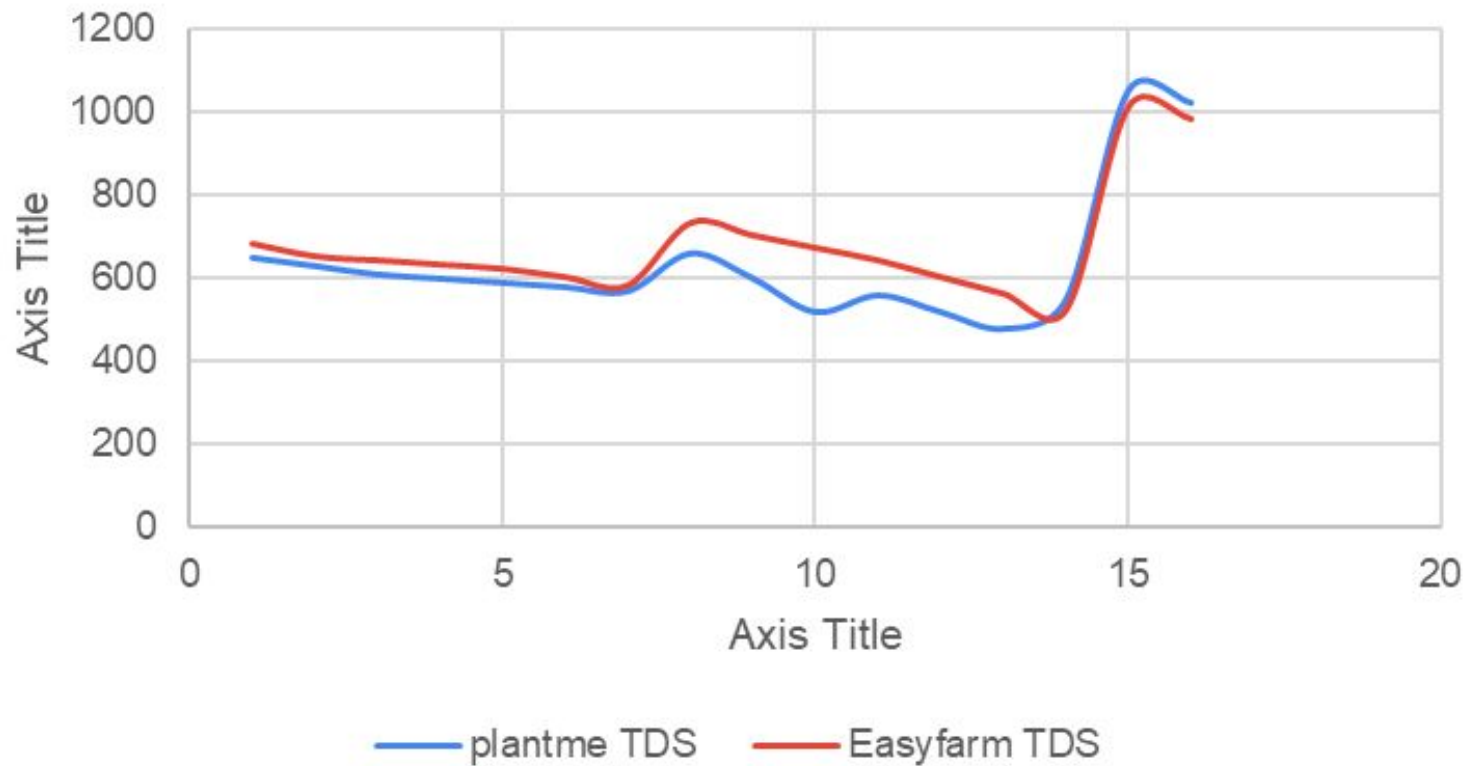
1.2 Crop Growth

- The study showed a noticeable difference in the impact of growth parameters on plants grown on different systems.
- Design of system, flow of water, had a clear influence on the crop growth.

Parameters	Vertical barrel system	A-Frame system
Shoot length	23 cm	26 cm
Plant spread	High with more leaves	High with more shoot
Root length	Moderate	Moderate
Root spread	Long and slender	Short and wide



TDS vs No. of days



Findings from the results were;

1. Water consumption in vertical barrel system was comparatively low with the A- frame system.

Reasons for high water consumption in A- frame system;-

- Larger surface area for water to evaporate.
- Roots being immersed in water consume more water.

Reasons for low water consumption in vertical barrel system:-

- Nutrients supplied by flowing water continuously as drops over roots of plants.
- No stagnant water surface present for evaporation.

2. Clogging of nutrient solution found in A-frame system due to wider roots.

https://drive.google.com/file/d/1uKu3InhyOfB_7mPrOdnOE9xToR8eP9fW/view?usp=sharing

3. Smaller diameter of the vertical barrel system caused lack of space for the plants to grow and flourish.
4. Disease detection was found to be more in A-frame system due to the stagnant water present.
5. Cleaning of vertical barrel system after one crop cycle found to be more easier than A-frame system.

CONCLUSION & FUTURE SCOPE

- Water can reach all the plants in vertical barrel systems, whereas in A frame system some plants may not get water due to clogging of pipe by the roots.
- Water consumption in vertical barrel system was comparatively lower than the A-frame system.
- Cleaning of vertical barrel system was easier than cleaning of A- frame system
- Smaller diameter of the vertical barrel system caused lack of space for the plants to grow and flourish.
- Vertical barrel systems can be easily converted to aeroponic systems.
- Development of nutrient solution.
- Locking mechanism for pods can be implemented to make it stable.
- Issue with power outage.
- Determining suitable plant for the vertical system.

ACHIEVEMENTS

- **First place in Project competition**
 - Conducted by The institution of Engineers, India and AISAT
 - Cash Prize of 10,000 INR
 - https://drive.google.com/file/d/1ozG9kbXjQB2CoG6aCGt8VNFs_wJqeiFG/view?usp=sharing
- **Participated in ASME Old Guard Oral Presentation**
 - ASME E-Fest 2021
- **Courses Completed - Inzamam**
 - Solidworks
 - <https://drive.google.com/drive/folders/1KBJSVNl3BfLEj3mbrvanHn1I1lB0q3N4>
 - Excel
 - https://drive.google.com/drive/folders/1R14Gxy2liiDXXF8g_mY21NGLs8ADnW-Z

ACHIEVEMENTS

- Courses Completed - Jen
 - <https://drive.google.com/drive/folders/1e3f70VTBHnMbx1wcG-6oxO-5NoXOBZTe>
- Courses Completed - Jayakrishnan G
 - https://drive.google.com/file/d/1gCVUe3_0NanTvnG3QoD3m_krCPoLf3di/view?usp=sharing

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- [1] Majid, M., Khan, J.N., Shah, Q.M.A., Masoodi, K.Z., Afroza, B. and Parvaze, S., 2021. Evaluation of hydroponic systems for the cultivation of Lettuce (*Lactuca sativa* L., var. *Longifolia*) and comparison with protected soil-based cultivation. *Agricultural Water Management*, 245, p.106572.
- [2] Kling, A., Pashek, V., Kumpan, V. and Konoplyov, Y., 2020, January. Modern Practices for the Cultivation of Leaf Vegetables in Hydroponics. In *International Scientific Conference The Fifth Technological Order: Prospects for the Development and Modernization of the Russian Agro-Industrial Sector (TFTS 2019)* (pp. 282-285). Atlantis Press.
- [3] Touliatos, D., Dodd, I.C. and McAinsh, M., 2016. Vertical farming increases lettuce yield per unit area compared to conventional horizontal hydroponics. *Food and energy security*, 5(3), pp.184-191.
- [4] Li, Q., Li, X., Tang, B. and Gu, M., 2018. Growth responses and root characteristics of lettuce grown in aeroponics, hydroponics, and substrate culture. *Horticulturae*, 4(4), p.35.
- [5] Heredia, N.A., 2014. Design Construction and Evaluation of a Vertical Hydroponic Tower.
- [6] Jat, G.S., Tomar, B.S., Lyngdoh, Y.A. and Saha, P., Vertical Farming of Leafy Vegetables under Protected Conditions.
- [7] Naskoori, K., Reddy, K.K., Reddy, M.V. and Devi, M.R., 2021. To study the scope of vertical farming in India: A.

The slide features a light beige background with a subtle paper-like texture. In the corners, there are delicate watercolor-style illustrations: top-left shows green leaves; top-right shows blue-green circular berries; bottom-left shows green leaves with small pink flowers; and bottom-right shows green leaves. Scattered across the background are numerous small, light blue circular speckles of varying sizes.

Thank You